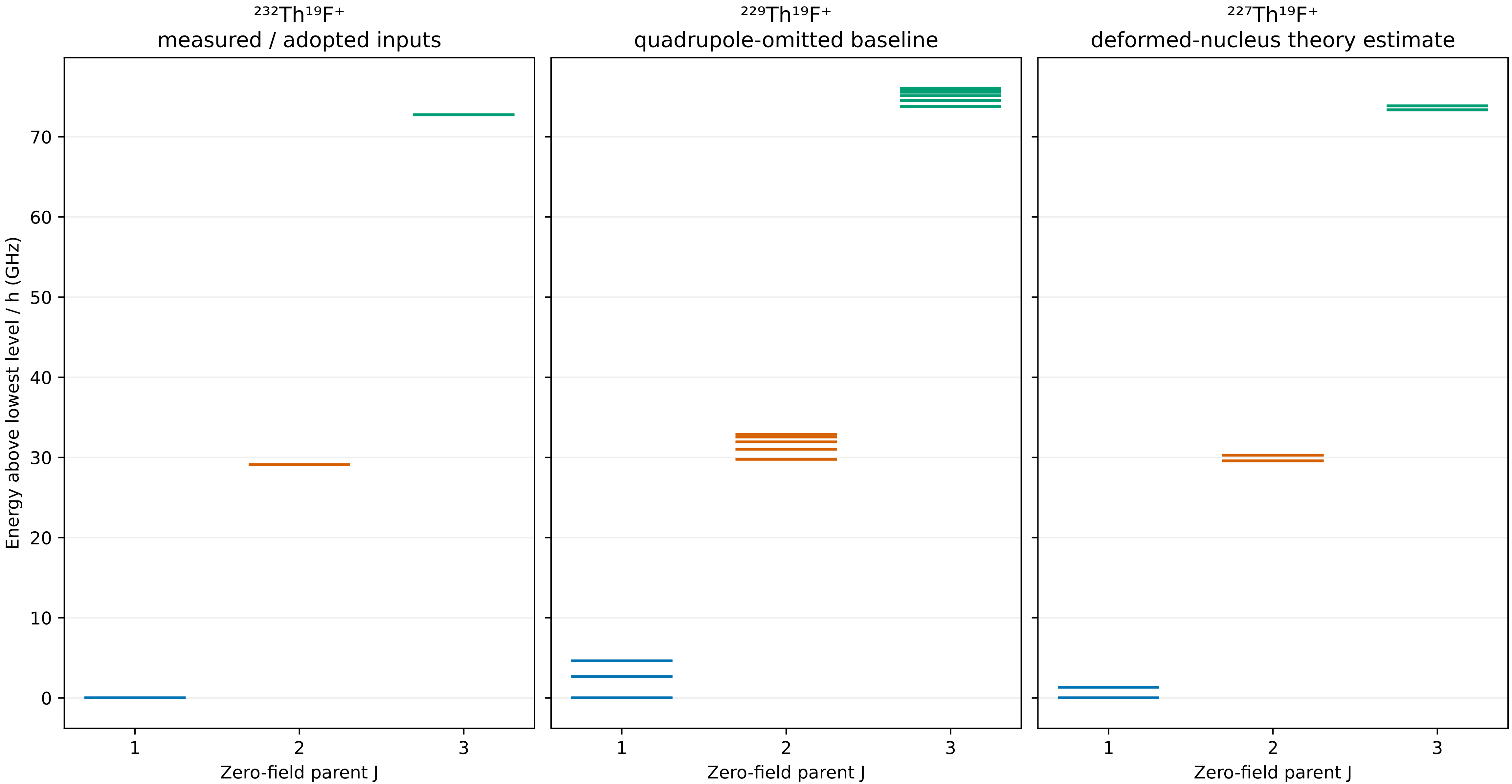
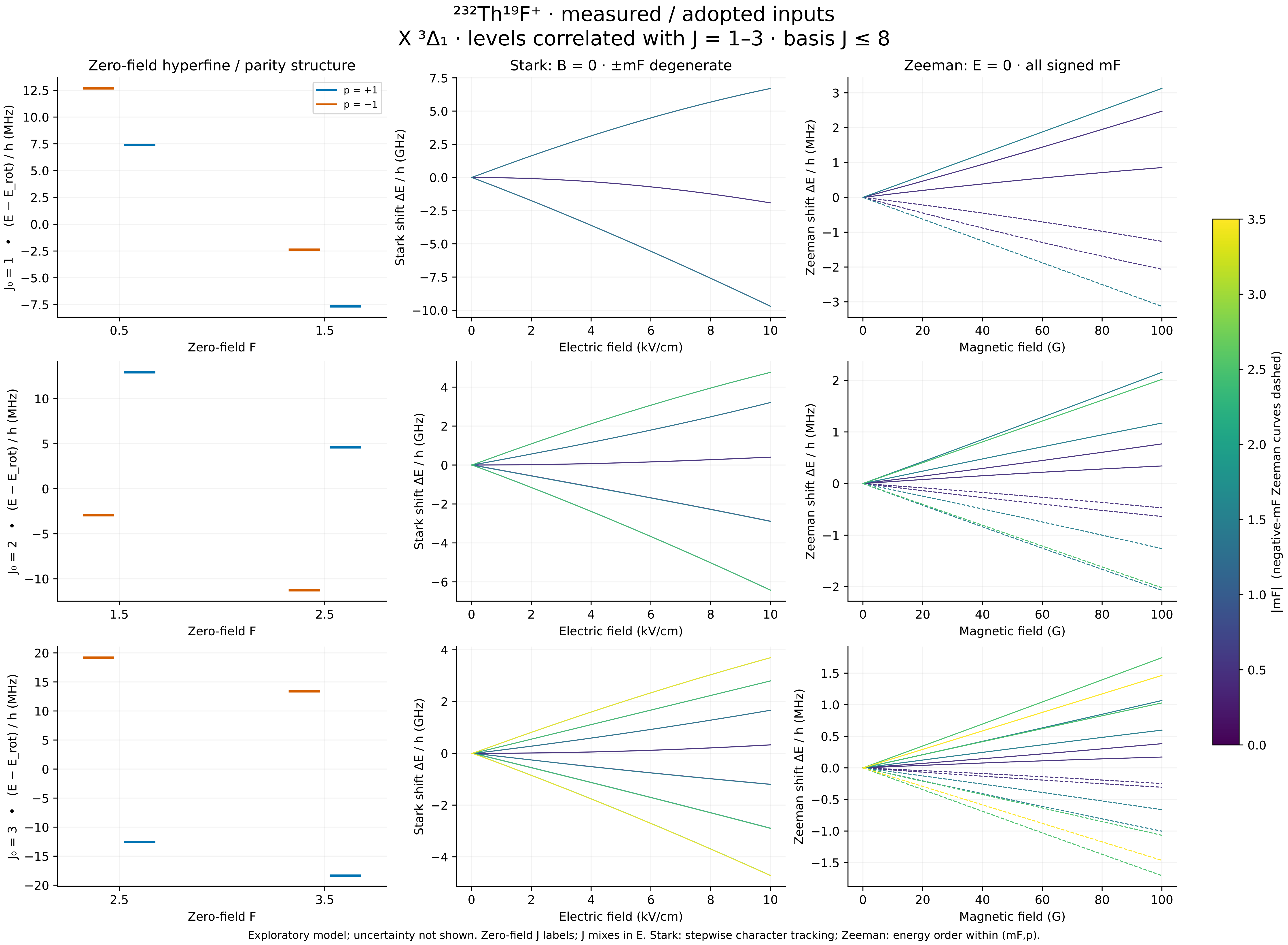
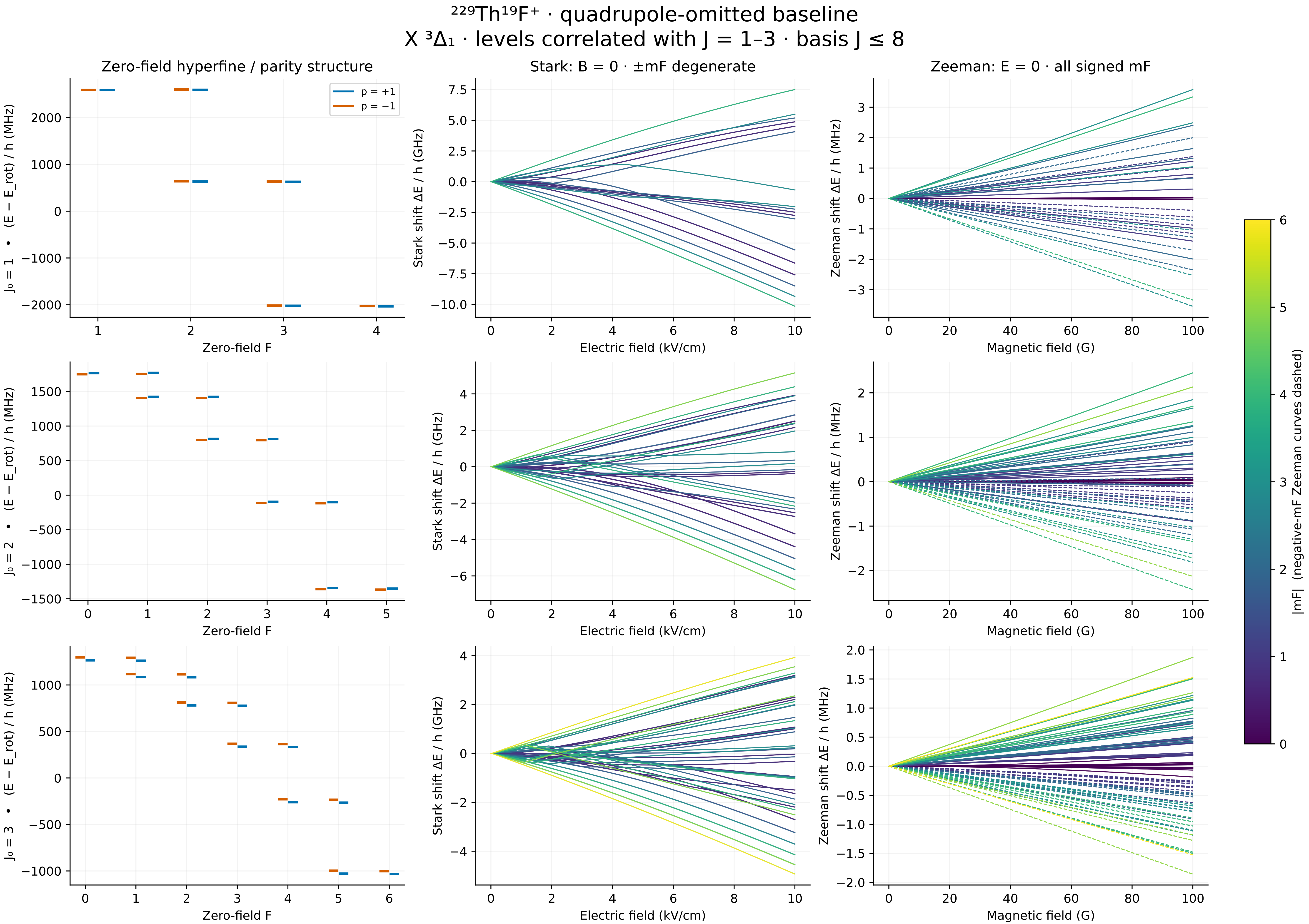
ThF^+ X $^3\Delta_1$ · zero-field rotational structure



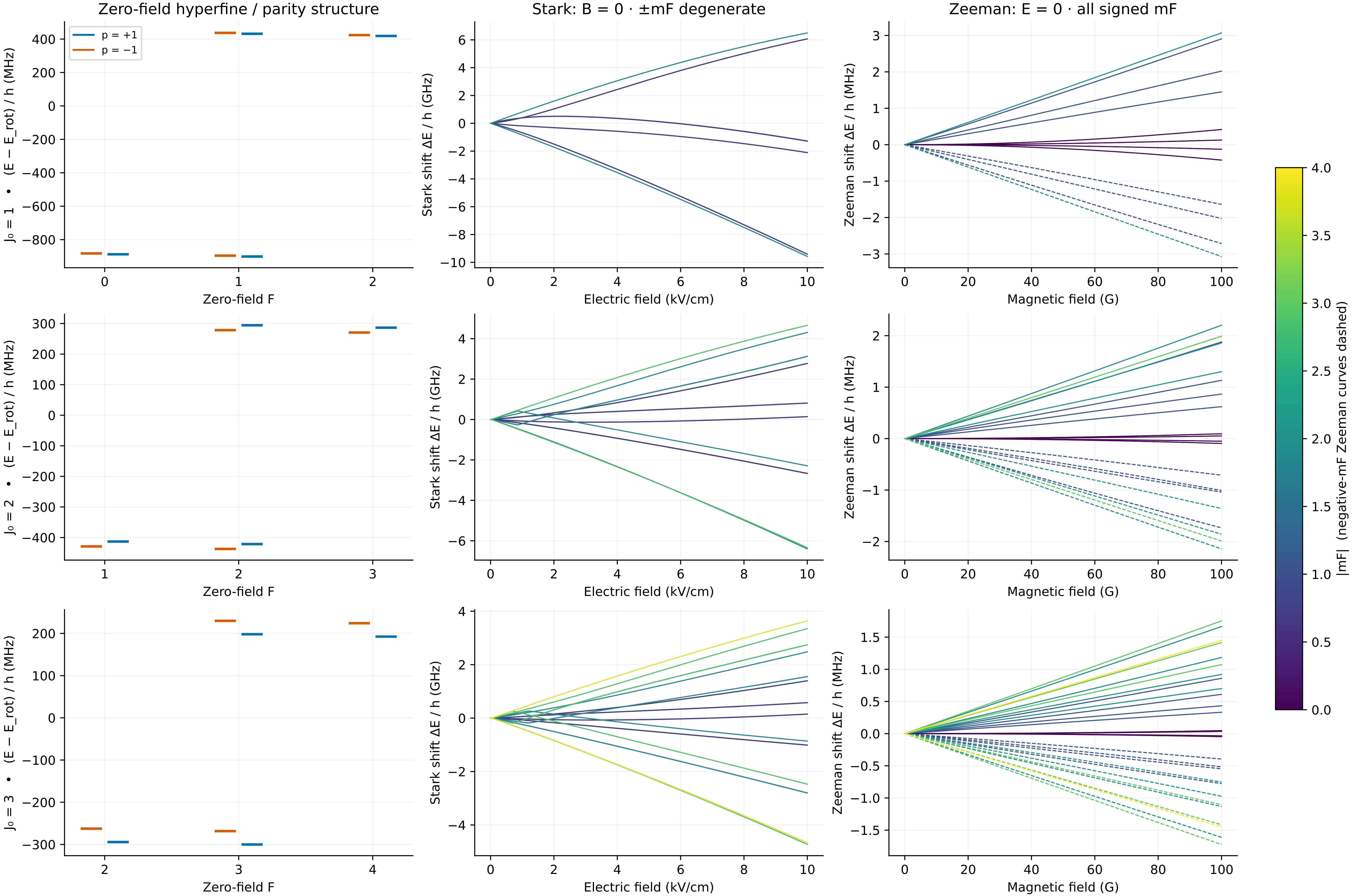
Each isotope has its own zero. Odd-isotope molecular constants are unscaled transfers from $^{232}ThF^+$.
Unknown Th spin rotation omitted; ^{229}Th quadrupole omitted. Exploratory models without physical uncertainty bands.


²³²Th¹⁹F⁺ · measured / adopted inputs
X ³Δ₁ · levels correlated with J = 1–3 · basis J ≤ 8
Zero-field hyperfine / parity structure
Stark: B = 0 · ±mF degenerate
Zeeman: E = 0 · all signed mF
p = +1
p = −1
J₀ = 1 • (E − E_rot) / h (MHz)
J₀ = 2 • (E − E_rot) / h (MHz)
J₀ = 3 • (E − E_rot) / h (MHz)
Zero-field F
Stark shift ΔE / h (GHz)
Electric field (kV/cm)
Zeeman shift ΔE / h (MHz)
Magnetic field (G)
|mF| (negative-mF Zeeman curves dashed)
Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).



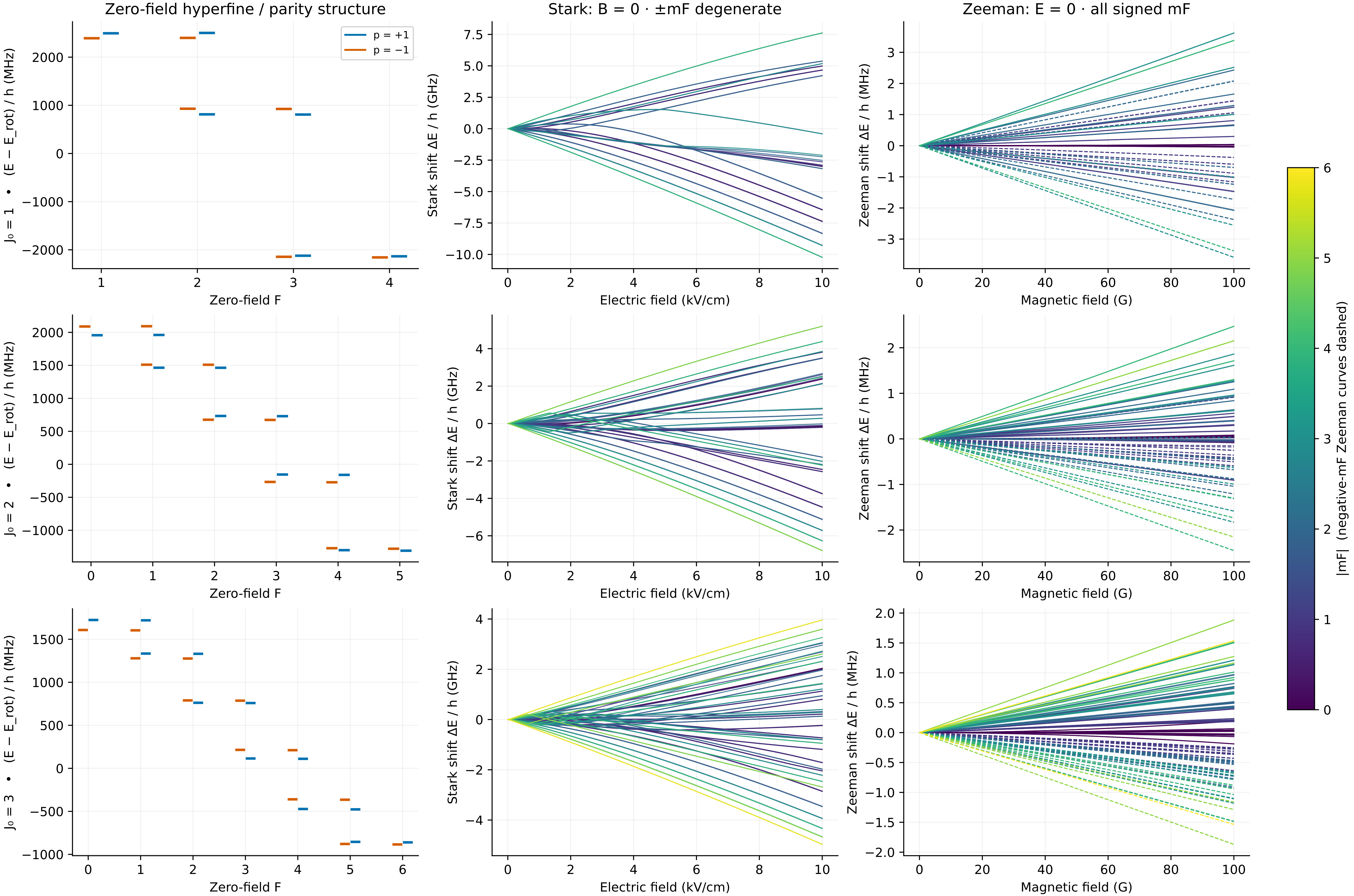
Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).
Molecular constants transferred unscaled from $^{232}ThF^{+}$; unknown Th spin rotation omitted; no physical uncertainty band.

$^{227}Th^{19}F^{+}$ · deformed-nucleus theory estimate
X $^3\Delta_1$ · levels correlated with J = 1–3 · basis J ≤ 8



Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).
Molecular constants transferred unscaled from $^{232}ThF^{+}$; unknown Th spin rotation omitted; no physical uncertainty band.

$^{229}Th^{19}F^+$ · UNVALIDATED quadrupole sensitivity case
X $^3\Delta_1$ · levels correlated with J = 1–3 · basis J ≤ 8



Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).
Molecular constants transferred unscaled from $^{232}ThF^+$; unknown Th spin rotation omitted; no physical uncertainty band.